

D 120185

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Name.....

Reg. No.....

SIXTH SEMESTER (CBCSS—UG) DEGREE EXAMINATION, MARCH 2025

Mathematics

MTS 6B 13—DIFFERENTIAL EQUATIONS

(2019 Admission onwards)

Time : Two Hours and a Half

Maximum : 80 Marks

Section A*Answer any number of questions.**Each question carries 2 marks.**Ceiling is 25.*

1. Give an example of a second order linear differential equation.
2. Find the general solution of the differential equation $\frac{dy}{dt} - 2y = 4 - t$.
3. State existence uniqueness theorem for first order linear equations.
4. Test exactness for the differential equation $(y \cos x + 2xe^y) + (\sin x + x^2e^y - 1)y' = 0$.
5. Find the solution of the initial value problem $y'' + 5y' + 6y = 0$, $y(0) = 2$, $y'(0) = 3$.
6. Show that $y_1 = e^{-t}$ and $y_2 = e^{2t}$ form a fundamental set of solutions of the equation $y'' - y' - 2y = 0$.
7. Find the general solution of $16y'' - 8y' + 145y = 0$.
8. Find a particular solution of $y'' - 3y' - 4y = 3e^{2t}$.
9. Determine a lower bound for the radius of convergence of series of solution of the differential equation $(1 + x^2)y'' + 2xy' + 4x^2y = 0$ about the point $x = 0$.
10. Find the Laplace transform of $f(t) = 5e^{-2t} - 3 \sin(4t)$, $t \geq 0$.

Turn over

11. Find the inverse Laplace transform of $F(s) = \frac{2s-3}{s^2-4}$.
12. State convolution theorem.
13. Define fundamental period. Find the fundamental period of $\sin\left(\frac{\pi x}{L}\right)$.
14. Define Fourier cosine series.
15. Give an example of a function which is neither odd nor even.

Section B

Answer any number of questions.

Each question carries 5 marks.

Ceiling is 35.

16. Solve $\frac{dy}{dx} = \frac{4y-3x}{2x-y}$.
17. Find the value of b for which $(ye^{2xy} + x) + bxe^{2xy}y' = 0$ is exact, and then solve it using that value of b .
18. Find the wronskian of two solutions of $(\cos t)y'' + (\sin t)y' - ty = 0$ without solving the equation.
19. Use the method of reduction of order to find a second solution of $t^2y'' - 4ty' + 6y = 0, t > 0$ whose one solution is $y_1(t) = t^2$.
20. Find the general solution of $y'' + y' - 6y = 12e^{3t} + 12e^{-2t}$.
21. Using Laplace transforms find the solution of the initial value problem $y'' + y = \sin(2t), y(0) = 2, y'(0) = 1$.
22. Solve the boundary value problem $y'' + y = 0, y(0) = 1, y(\pi) = a$, where a is a given number.
23. Find the Fourier series for $f(x) = \begin{cases} x & \text{if } 0 \leq x < 1 \\ 1 & \text{if } 1 \leq x < 2 \end{cases}, f(x+4) = f(x)$.

Section C

*Answer any **two** questions.*

Each question carries 10 marks.

Maximum 20 marks.

24. Solve the initial value problem $y' = 2t(1 + y)$, $y(0) = 0$ by the method of successive approximations.
25. Find a solution of $y'' - xy = 0$, $-\infty < x < \infty$ in powers of $(x - 1)$.
26. Find the solution of $y'' + 2y' + 2y = \delta(t - \pi)$; $y(0) = 1$, $y'(0) = 0$.
27. Find the Fourier series of $f(x) = \begin{cases} x + 1 & \text{if } -1 \leq x < 0 \\ 1 - x & \text{if } 0 \leq x < 1 \end{cases}$; $f(x + 2) = f(x)$.